

The de Sitter–Unruh Acceleration Scale: A Geometric Reframing of the MOND Constant a_0 , and No-Go Theorems for Its Modified-Inertia Completion

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Abstract

The dynamics of galaxies require, below a characteristic acceleration $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$, either unseen matter or a modification of the law of gravity/inertia (Milgrom 1983). This scale numerically coincides with the cosmological quantity $c\sqrt{\Lambda}$, where Λ is the cosmological constant. This paper does three things, and is careful to keep them separate.

1. **A reframing (modest, not original).** The same number is written as a *surface gravity of the cosmological horizon*: $a_0 = c^2 \sqrt{(\Lambda/32\pi)} = (c/2) \sqrt{(G\rho_\Lambda)} = cH_\Lambda/Z$, with $Z = \sqrt{(32\pi/3)} \approx 5.789$. This is a few lines of algebra; anyone can check it in minutes.
2. **A mechanism for the *shape* (known physics).** The square-root “deep-MOND” law, and with it the Radial Acceleration Relation (RAR) and the Baryonic Tully–Fisher Relation (BTFR), follow from treating inertia as a response to the de Sitter–Unruh temperature of the cosmic horizon (Milgrom 1999; Deser & Levin 1997). The mechanism is real; it is not new here.
3. **Rigorous limits (the genuinely new content).** We prove, with explicit calculations, exactly what this picture *cannot* do. (i) The $O(1)$ coefficient Z is **not derived** — it is a free choice. (ii) The “temperature route” that gives the shape predicts an acceleration scale $\sim 12\times$ too large, and at the level of a real action it produces the **wrong sign** (it raises inertia, i.e. anti-MOND): we state this as a passivity theorem. (iii) **No covariant modified-inertia completion exists** in three exhaustive cases — local, field-theoretic, and nonlocal — each blocked by a named obstruction (an Ostrogradsky ghost, a Cassini-violating metric coupling, and the passivity sign theorem respectively).

The honest conclusion: $a_0 \sim c\sqrt{\Lambda}$ is a **forced scale** — **but the MOND sign and the precise normalization are *postulated*, not derived**. This is a suggestive geometric/thermodynamic way to look at the MOND constant, with sharply proven boundaries. It is **not** a complete theory of MOND, and it is **not** a theory of everything. The one observationally distinctive consequence — a *declining* acceleration scale, $a_0(z=3) \approx 0.74 a_0(0)$, tracking the dark-energy density — is falsifiable this decade.

This paper supersedes and corrects, in the direction of *less* claimed, an earlier and over-stated description of this material.

1. Introduction and background

1.1 The problem MOND addresses

A spiral galaxy rotates. If you measure how fast stars and gas orbit at a radius r , Newtonian gravity predicts the orbital speed v from the visible (baryonic) mass M inside that radius: $v^2 = GM/r$, where G is Newton's constant. The prediction fails — observed speeds stay roughly *constant* out to large radii (“flat rotation curves”) instead of falling off. The two standard responses are (a) **dark matter**: posit extra invisible mass, or (b) **MOND** (MOdified Newtonian Dynamics, Milgrom 1983): posit that the law itself changes when accelerations get very small.

MOND's empirical core is a single new constant of nature, the **acceleration scale a_0** . Above a_0 , physics is Newtonian. Below a_0 (the “deep-MOND” regime, where galaxy outskirts live), the effective gravitational acceleration g is no longer the Newtonian value $g_N = GM/r^2$ but instead

$$g = \sqrt{a_0 \cdot g_N} \text{ (deep-MOND limit). (1.1)}$$

Because $g = v^2/r$ for a circular orbit, equation (1.1) gives $v^4 = G M a_0$ — a flat rotation curve whose speed depends only on baryonic mass. Empirically $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$, and this one number, with one stellar mass-to-light ratio, reproduces the rotation curves of ~ 175 galaxies (the SPARC sample; Lelli, McGaugh & Schombert 2016) to within their scatter.

1.2 The coincidence this paper is about

The number a_0 is suspiciously close to two *cosmological* quantities:

- cH_0 — the speed of light times the Hubble constant (the expansion rate of the universe). Numerically $cH_0 \approx 6.8 \times 10^{-10} \text{ m s}^{-2}$, so $a_0 \approx cH_0/6$.
- $c\sqrt{\Lambda}$ — where Λ (the cosmological constant) is the tiny constant energy density of empty space driving cosmic acceleration. Numerically $a_0 \approx c\sqrt{\Lambda} \times O(1)$.

In standard Λ CDM cosmology this coincidence is treated as **accidental** — a_0 is a galaxy-scale fact, Λ is a cosmic-scale fact, and their proximity is chance. The premise explored here is the opposite: that the coincidence is **causal** — the galaxy acceleration scale is *set by* the cosmological constant. We will see that this premise is *partly* defensible (the scale is forced; the shape has a mechanism) and *partly* not (the exact coefficient and the sign are not derived). Stating which is which, rigorously, is the point of the paper.

1.3 What is claimed, and what is not

To avoid the trap this material previously fell into, here is the ledger up front.

Statement	Status
a_0 has the <i>order of magnitude</i> of a cosmological acceleration ($c\sqrt{\Lambda} \sim cH_0$)	Forced (Sec. 2–3)
The deep-MOND $\sqrt{\cdot}$ -law, RAR, BTFR	Derived from de Sitter–Unruh inertia (known physics, Sec. 3)
The <i>exact value</i> $a_0 = c^2\sqrt{(\Lambda/32\pi)}$	Not derived — one $O(1)$ coefficient is a free choice (Sec. 4.1)
The MOND <i>sign</i> (inertia <i>lowered</i> at small a) from the cosmic vacuum	Not derived — provably so (Sec. 4.2)
A covariant theory underlying all this	Does not exist in three exhausted cases (Sec. 4.2)
A theory of everything / the Standard Model	Not claimed. Disavowed.

2. The acceleration scale as a horizon surface gravity

2.1 The quantities involved (background)

- **The cosmological constant Λ .** Empty space has a small constant energy density. In Einstein’s equations this appears as Λ , with units of $1/\text{length}^2$. Measured value $\Lambda \approx 1.1 \times 10^{-52} \text{ m}^{-2}$. Equivalently it is a *dark-energy mass density* $\rho_\Lambda = \Lambda c^2/(8\pi G) \approx 5.8 \times 10^{-27} \text{ kg m}^{-3}$ (this is just the definition that turns the geometric Λ into a density, using G and c ; the 8π is the standard Einstein-equation factor).
- **The de Sitter rate H_Λ .** A universe containing only this dark energy expands exponentially at a rate $H_\Lambda = c\sqrt{(\Lambda/3)}$ (the factor 3 is the Friedmann equation, $H^2 = 8\pi G\rho/3$, specialized to $\rho = \rho_\Lambda$). H_Λ is close to today’s Hubble constant H_0 because dark energy dominates today.
- **Surface gravity.** A horizon of radius R has a characteristic acceleration, its *surface gravity* $\kappa_{\text{surf}} = c^2/(2R)$ — the same formula that gives a black hole’s surface gravity (there R is the Schwarzschild radius). It is “how hard” the horizon pulls at its own scale.

2.2 The algebra (anyone can verify this)

Start from the hypothesis that the dark-energy density sets the scale through a surface-gravity-like relation,

$$a_0 = (c/2)\sqrt{(G \rho_\Lambda)}. \quad (2.1)$$

The $(c/2)$ is a choice of $O(1)$ coefficient — call its origin $\kappa = 1/2$; we will return to the fact that it is not forced (Sec. 4.1). Substitute $\rho_\Lambda = \Lambda c^2/(8\pi G)$:

$$a_0 = (c/2)\sqrt{(G \cdot \Lambda c^2/8\pi G)} = (c/2) \cdot c\sqrt{(\Lambda/8\pi)} = c^2\sqrt{(\Lambda/32\pi)}. \quad (2.2)$$

Numerically, with the measured Λ , this gives $\mathbf{a}_0 = 9.36 \times 10^{-11} \text{ m s}^{-2}$ — within ~20% of the empirical MOND scale (and consistent with it once the stellar mass-to-light ratio and the choice of interpolation function are accounted for; this paper does not re-litigate that fit).

Rewrite (2.2) as a surface gravity, $a_0 = c^2/(2R^*)$:

$$R^* = c/\sqrt{G\rho_\Lambda} = \sqrt{(8\pi/\Lambda)}. \quad (2.3)$$

So a_0 is *exactly* the surface gravity of a horizon of radius R^* , the natural free-fall radius of a universe at the dark-energy density. Finally, dividing (2.2) by $H_\Lambda = c\sqrt{\Lambda/3}$:

$$a_0 = c H_\Lambda / Z, \quad Z = \sqrt{(32\pi/3)} \approx 5.789. \quad (2.4)$$

Equations (2.1)–(2.4) are the entire “reframing.” They are algebraic identities given the one input (2.1); a referee can check them in minutes. **Read this way, a_0 is the acceleration associated with the cosmological horizon — galaxies in the deep-MOND regime are systems whose internal accelerations have dropped to the level set by the universe’s own boundary.** That is the picture. Whether it is a *theory* depends on the next two sections.

3. The shape: deep-MOND from de Sitter–Unruh modified inertia

This section recovers the *functional form* (1.1) — the square-root law, the RAR, the BTFR — from a physical mechanism. The mechanism is **modified inertia from the de Sitter–Unruh temperature** (Milgrom 1999; the temperature structure is Deser & Levin 1997). It is established physics; nothing here is original. It matters because it is what separates “an interesting numerical coincidence” from “a mechanism for the law.”

3.1 Background: the Unruh and de Sitter–Unruh temperatures

The **Unruh effect** (1976): an observer accelerating through the quantum vacuum with proper acceleration a perceives the vacuum as a thermal bath at temperature $T_U = \hbar a/(2\pi c k_B)$. Acceleration and temperature are proportional; an accelerated detector “feels warm.”

In de Sitter space (a universe dominated by Λ) there is, additionally, the **Gibbons–Hawking temperature** of the cosmological horizon, $T_{dS} \propto \hbar H/k_B$ — empty accelerating space is itself warm. An observer who is *also* accelerating sees a combination. The combined (Deser–Levin) temperature for acceleration a in de Sitter space is

$$T(\mathbf{a}) \propto \sqrt{(\mathbf{a}^2 + (cH)^2)}. \quad (3.1)$$

(The accelerations add in quadrature, like the magnitudes of perpendicular vectors, because the two temperatures come from independent horizons.)

3.2 The deep-MOND limit

Milgrom's modified-inertia proposal: a body's inertia responds not to its acceleration directly, but to the *excess* temperature it feels over the cosmic background — i.e. to

$$\mathbf{T}_{\text{eff}}(\mathbf{a}) = \mathbf{T}(\mathbf{a}) - \mathbf{T}(\mathbf{0}) \propto \sqrt{(\mathbf{a}^2 + (c\mathbf{H})^2)} - c\mathbf{H}. \quad (3.2)$$

Now take the **low-acceleration limit** $a \ll cH$ (galaxy outskirts). Expand the square root: $\sqrt{(\mathbf{a}^2 + (c\mathbf{H})^2)} = cH\sqrt{1 + \mathbf{a}^2/(c\mathbf{H})^2} \approx cH(1 + \mathbf{a}^2/(2(c\mathbf{H})^2)) = cH + \mathbf{a}^2/(2c\mathbf{H})$. Subtracting cH ,

$$\mathbf{T}_{\text{eff}} \approx \mathbf{a}^2/(2c\mathbf{H}) \text{ (deep-MOND)}. \quad (3.3)$$

The effective inertial response is *quadratic* in the acceleration when accelerations are small. If we identify the Newtonian source $\mathbf{g}_N$ with this effective response (the body's true dynamical acceleration $\mathbf{a}$ must produce the Newtonian pull through the *reduced* inertia), the operative relation is $\mathbf{a}^2 \propto \mathbf{g}_N \cdot (c\mathbf{H})$, i.e.

$$\mathbf{a} = \sqrt{(\mathbf{a}_0 \mathbf{g}_N)}, \text{ with } \mathbf{a}_0 \propto c\mathbf{H}. \quad (3.4)$$

This is the deep-MOND law (1.1). From it follow immediately:

- the **RAR** (the observed-vs-baryonic acceleration relation), via the full interpolation $\mu(a) = a / (\sqrt{(\mathbf{a}^2 + \mathbf{a}_H^2)} - \mathbf{a}_H)$ that smoothly joins Newtonian ($a \gg a_0$) to deep-MOND ($a \ll a_0$);
- the **BTFR** $v^4 = G M a_0$, the tight relation between a galaxy's baryonic mass and its flat rotation speed.

We verified (1.1), the RAR interpolation, and $v^4 = G M a_0$ symbolically; they are exact consequences of (3.2). **So the *shape* of MOND genuinely follows from de Sitter–Unruh inertia.** This is the strongest part of the picture, and — per §1.3 — the part most worth a relativist's attention.

4. The limits, proven

Sections 2–3 are the optimistic half. This section is the half that keeps the work honest: precisely what cannot be derived, with the obstruction named in each case. **This is the new content of the paper.**

4.1 The coefficient is not derived; the two routes disagree

The scale a_0 emerged with a free $O(1)$ coefficient — the $\kappa = \frac{1}{2}$ in (2.1), equivalently the precise value of Z in (2.4). Nothing above fixes it. We separately verified that κ cannot be forced by the usual candidate principles (ghost-freedom, unitarity, holographic entropy bounds): each of those probes only *signs, ratios, or the overall scale*, and κ is an overall normalization invisible to them. **So a_0 's order of magnitude is forced (it is a cosmological acceleration), but its exact value is a one-parameter choice, not a zero-parameter prediction.** That is a completely standard situation in physics — a derived functional form with one undetermined $O(1)$ constant — and it is stated here as such rather than disguised.

There is a sharper problem, and it is the most important physical statement in the paper. **The route that gives the *shape* (Sec. 3) and the route that gives the *scale* (Sec. 2) are not the same theory.** The shape comes from a *temperature*: the natural acceleration scale of the de Sitter–Unruh temperature is $\sim cH_\Lambda$ (even $\sim 2cH_\Lambda$ for the static-patch reading), i.e. it predicts a_0 of order cH_Λ . The observed value (2.4) is $a_0 = cH_\Lambda/Z \approx cH_\Lambda/5.8$ — about an **order of magnitude smaller** than the temperature route predicts (the ratio $2Z \approx 11.6$ is the “ $\sim 12\times$ overshoot”). The reason is structural and worth stating clearly:

Every de Sitter temperature is *linear* in H (an energy scale, $\propto cH$), while $a_0 = c^2\sqrt{(\Lambda/32\pi)}$ is a *square root of a density* ($\propto \sqrt{\rho_\Lambda}$). A linear-in- H temperature can never equal a square-root-of-density acceleration without an extra ingredient. The 32π in Z is a *gravitational* normalization (the 8π of Einstein’s equations, inside $\rho_\Lambda = \Lambda c^2/8\pi G$), and it is invisible to a kinematic temperature.

So the horizon interpretation is genuinely “onto something” for the *shape*, but the *scale* it naturally predicts is wrong by $\sim 12\times$, and welding the temperature-route shape to the density-route scale is an unsolved problem — not a completed one.

4.2 No covariant modified-inertia completion exists (a three-horn no-go)

A “reframing” becomes a “theory” when it has a **covariant action** — a Lorentz-invariant Lagrangian that (a) yields the deep-MOND law, (b) recovers Newton/GR at high acceleration, (c) is *stable* (no runaway/ghost), and (d) survives Solar-System tests. There are exactly three ways to build modified inertia, and we find **all three are blocked**, each by a named, calculable obstruction.

(i) Local route $\rightarrow$ Ostrogradsky ghost. The natural local construction gates a particle’s inertia by its acceleration magnitude $|a|$. But $|a|$ is a *second time derivative* of position, so the Lagrangian depends on $\ddot{x}$. *Background: Ostrogradsky’s theorem* states that any non-degenerate Lagrangian containing second (or higher) time derivatives has a Hamiltonian that is *unbounded below* — there is a “ghost,” a mode whose energy can go to $-\infty$, making the theory unstable. We verified the gate is non-degenerate ($d^2L/d(\ddot{x})^2 \neq 0$) and exhibits the ghost explicitly (a propagator pole with negative residue). Adding a preferred-frame (“aether”) vector does not help: the vector *labels* the cosmic rest frame but cannot lower the derivative order of the gate. This is precisely Milgrom’s 1994 modified-inertia no-go, made explicit.

(ii) Field route $\rightarrow$ modified gravity, fails the Solar System. Promote the modification to a dynamical field that sits in the *metric* sector. Then it is no longer modified *inertia* but modified *gravity*, and it generically moves the gravitational potential Φ . *Background:* in any theory built from a metric, energy–momentum conservation (a consequence of coordinate invariance, the “Bianchi identity”) *links* the different components of the gravitational field. We show that the kind of anisotropic (“traceless shear”) stress such a field produces has a non-zero divergence, which conservation can only balance by sourcing a pressure that *feeds back into* Φ . A theory that changes Φ at the relevant level is excluded by **Cassini**’s measurement of the post-Newtonian parameter

γ ($|\gamma - 1| < 2 \times 10^{-5}$, Bertotti, Iess & Tortora 2003). So the field route survives stability only by becoming the thing the data already forbids.

(iii) Nonlocal route $\rightarrow$ the passivity sign theorem. The one remaining corner is a *fully nonlocal* action — inertia as a memory response of the particle’s worldline to the cosmic vacuum (Milgrom 1999). This is ghost-free (a nonlocal “form factor” can have a single healthy pole, evading Ostrogradsky). It is the most physically natural route and the closest the framework comes to a real completion. But it fails for one sharp, provable reason, the central technical result of this paper:

The passivity $\rightarrow$ anti-MOND sign theorem. Let a particle acquire its inertial correction from coupling to *any* cosmological/de Sitter source that is (causal) — its response is analytic in the upper half frequency-plane (Kramers–Kronig) — and (ghost-free/unitary) — its spectral density satisfies $\rho(\omega) \geq 0$ (Källén–Lehmann positivity). Then the change in inertia at low acceleration is $\delta m = 2 \int_0^\infty \rho(\omega')/\omega'^2 d\omega' \geq 0$ — inertia is **raised**. MOND requires inertia *lowered* ($\delta m < 0$), which demands $\rho(\omega) < 0$ over some band — a negative-norm ghost. Hence “active (sign-flipping) + causal + ghost-free + MOND-signed” is **overdetermined and impossible** for any passive, unitary, stationary vacuum.

Background on what makes this tight. A “passive” medium is one in thermal equilibrium that can only *absorb* net energy from a probe (dissipation); the fluctuation–dissipation theorem ties its noise to that positive dissipation, and the sign of the low-frequency inertia shift follows. To *lower* inertia you need an *active* (gain) medium — one that does net work *on* the probe, like a laser medium. We checked the candidate cosmological sources for activity and found every realizable one is passive: the expanding background only reshapes the *noise* (not the sign-bearing dissipation); the de Sitter (Bunch–Davies) vacuum restricted to a static patch is *exactly thermal* (the KMS condition) with detailed balance; horizon thermodynamics (Jacobson, Padmanabhan) obeys the second law with positive entropy production. The only genuinely active mechanism — a driven non-equilibrium steady state (“frenesy”) — is real physics but cannot be sourced from a de Sitter vacuum without postulating an external drive, and in any case operates at the wrong (super-horizon) frequencies.

Important both-ways clarification: the wall is **passivity (thermodynamics)**, *not* causality. Causality alone permits a MOND-signed kernel — we exhibit an explicit causal counterexample — but that kernel describes a *gain medium*, which the passive cosmic vacuum is not. This sharpens, rather than weakens, the result: the obstruction is precisely identified.

Net. All three horns are blocked: local = ghost, field = Cassini, nonlocal = unsourceable active kernel. **The MOND sign cannot be sourced from the passive de Sitter vacuum.** Therefore, within this picture, the MOND sign — and with it the specific normalization — must be *postulated*. $a_0 \sim \sqrt{\Lambda}$ is a forced scale; the dynamics that would make it a derived *theory* of inertia provably do not exist in the passive-vacuum framework.

5. Predictions and observational status

A reframing earns the word “predictive” only if it forecasts something that was not used to build it. Here is the honest breakdown.

Genuinely distinctive (differs from ordinary constant- a_0 MOND): - $a_0(z)$ **decline**. If a_0 tracks the dark-energy density, and dark energy dilutes (even slightly) with redshift on the DESI-preferred evolving branch, then a_0 *decreases* into the past: $a_0(z=3) \approx 0.74 a_0(0)$. Equivalently, the high-redshift BTFR zero-point sits slightly *below* the local one. This is the cleanest distinctive test, accessible to ELT/JWST/ALMA high- z kinematics this decade. **Caveat:** it is hostage to the dark-energy equation of state — if DESI consolidates on a pure cosmological constant ($w = -1$), this signal vanishes and the picture degenerates to ordinary MOND. - **A CMB-apex acceleration dipole** ($\approx 0.06\%$, fixed sky direction) — unique to a preferred-frame realization, but below every near-term experimental floor.

Modified-inertia vs modified-gravity discriminators (test that it is *inertia*, not gravity): the Solar-System $a_0/2$ channel (modified inertia evades Cassini where modified gravity is excluded); a cluster-member velocity-dispersion correlation with infall phase (zero for modified gravity).

Real but MOND-shared (test MOND vs dark matter, not this picture specifically): the RAR and its scatter, the external field effect (wide binaries, Gaia DR4), galaxy-cluster residuals, dwarf-galaxy dynamics, the weak-lensing RAR (Euclid DR1).

The honest sting: the *most distinctive* predictions are the hardest to access ($a_0(z)$ is hostage to w ; the CMB dipole is below floor), while the *accessible* ones are MOND-shared. The decisive near-term gate is the $a_0(z)$ / **high- z BTFR-sign** measurement.

6. Conclusion

This paper has tried to do the one thing speculative frameworks most often fail to do: **separate what is forced from what is chosen, and prove the difference.**

- **Forced:** a_0 has the magnitude of a cosmological acceleration; $a_0 \sim c\sqrt{\Lambda}$ is a horizon surface gravity (Sec. 2). The deep-MOND $\sqrt{\cdot}$ -law, the RAR, and the BTFR follow from de Sitter–Unruh modified inertia (Sec. 3). These are real, and the shape mechanism is the genuinely interesting physics.
- **Not forced, and provably so:** the exact coefficient Z (a free $O(1)$, Sec. 4.1); the agreement of the temperature route’s scale with the density route’s scale (they differ by $\sim 12\times$, Sec. 4.1); the MOND *sign* from the cosmic vacuum (forbidden by the passivity theorem, Sec. 4.2); a covariant completion (excluded in all three cases, Sec. 4.2).

The defensible position is therefore: $a_0 = c^2\sqrt{(\Lambda/32\pi)}$ is a **suggestive geometric/thermodynamic reframing of the MOND acceleration scale, with a real mechanism for its functional form and sharply proven limits on its completion. It is not a theory of MOND, and it is emphatically not a theory of everything.** The forward direction is observational: the declining- $a_0(z)$ signature is the test that could promote this from a reinterpretation to a falsified-or-confirmed physical claim within the decade.

We state these limits plainly because the alternative — pretending an undetermined coefficient was derived, or that a reframing was a theory — is how speculative work loses the right to be taken seriously. The boundaries proven here are, we think, more valuable than the coincidence that motivated them.

Acknowledgements and provenance

The de Sitter–Unruh modified-inertia route to MOND is due to M. Milgrom (1999); the de Sitter–Unruh temperature structure to S. Deser and O. Levin (1997). The empirical MOND relations used here are from the SPARC program (Lelli, McGaugh & Schombert). This manuscript intentionally claims *less* than, and corrects, an earlier description of the same material by the author. All symbolic and numerical checks (the algebra of Sec. 2, the deep-MOND expansion of Sec. 3, the Ostrogradsky, Bianchi, and passivity-sign calculations of Sec. 4) are reproducible from the public repository.

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